

Alexander Loker

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Education

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering – 3.7 GPA

2025 – Present

Waterloo, ON, Canada

Skills

Hardware: Altium, LTspice, SolidWorks, RasPi, AutoCAD

Software & Protocols: C++, Python, Git, Linux, PWM, DMX512, I2C, SPI, UART

Tools: Soldering, 3D Printing, Oscilloscope

Experience

WET Design

May – Aug 2026

Product Designer & Engineer

Sun Valley, CA, USA

- Built a handheld DMX light tester from the ground up to replace the **\$1200** Swisson XMT-500, bringing per unit cost to **\$300** in parts while adding fault protection and an externally programmable layer.
- Wired and tested an illuminated water and air feature that required lighting and pump control, drove both from a single DMX universe through an inhouse DMX to power supply converter.
- Owned the team's Weigl ProCommander HXi integration for a fountain rig, wired and documented the control box across DMX, analog and digital I/O, and trained 4 engineers on the platform.

Projects

DMX Handheld Light Tester – WET Design 🌐

Jun – Present

- Designed a custom **RasPi CM4** carrier board in **Altium** from schematic to PCB layout, integrating a WET RS-485 port, battery management, and power distribution.
- Safely integrated a **BQ charger IC**, **buck-boost IC**, and CM4 3.3V rail, using low-impedance routing and proper thermal balancing to carry up to **10W**.
- Writing the software in Python, building **DMX512** framing and timing on top of **UART** alongside **I2C** and **SPI** drivers for the BMS and LCD.
- Iterated the enclosure around software and PCB revisions while customizing UX/UI in **SolidWorks** and **PyGame** to accommodate a modular script, on-site safe, and one-handed design.

Companion Robot 🌐

Jan – Apr, Sep – Present

- Led end-to-end design of an autonomous companion robot spanning mechanical design (**SolidWorks**), electrical components and routing, and software implementation (**Python**).
- Replaced off-the-shelf parts with a custom battery management and power distribution PCB, verified in **WEBENCH** and designed in **Altium**, for reliable and safe power management.
- Designed **hardware and software** for a tri-omniwheel drivetrain on a **Pi Pico** with **Python PID** motor control and a tracking algorithm, MediaPipe, enabling **autonomous 360°** human-following.
- Compacted robot volume by **26%** while embedding servos, microphones, and a camera module directly into the shell, utilizing compact interlocking joints.

Under-voltage Lockout Circuit – Biotron Exoskeleton Team 🌐

Feb – Mar 2026

- Designed and verified a **UVLO** circuit schematic in **LTspice**, and transformed the schematic into a PCB in **Altium**.
- Created under-voltage lockout logic via **Schmitt trigger hysteresis** and controlled current flow through a **dual-MOSFET switch system** ensuring voltage lockout at **24.1V**.

Autonomous Air Hockey Defense Machine 🌐

Nov 2025

- Designed and assembled a **fully automated** puck-deflection robot intercepting **76 of 100** shots against live opponents.
- Integrated an **optical sensor** and coded the logic for real-time foul detection.
- Developed a real-time trigonometric **predictive algorithm in C++** to calculate puck landing position from sensor input whilst collecting and analyzing sensor data to validate system accuracy.